

REEFPRINT/KHANYA: ore-hardness prediction on the belt and sulphide characterisation in the laboratory.

An analogue-data proof of concept and staged validation plan for South African concentrators

Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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Abstract

South African concentrators set mill feed, grind and reagents before the laboratory can say what was in the ore. REEFPRINT/KHANYA is a decision-support system that gives the shift team that reading earlier, at two speeds, and leaves the decision with a person. On the belt, a short-wave infrared camera scores each 30-second parcel of mill feed for grinding hardness with a conservative upper bound, and Bond’s law turns the bound into a proposed feed rate inside limits set by the metallurgist. In the shift laboratory, a microscope model maps pyrrhotite, chalcopyrite and pentlandite in polished sections and reports their areas with grain-by-grain evidence. An operator approves, modifies or rejects every proposed change, the plant’s own control system keeps authority, and every step is signed and recorded. All results so far come from public analogue datasets and a plant simulator. On 146 public porphyry-ore composites the hardness model reached an out-of-fold R^2 of 0.479, its 90% upper bound covered 93.2% of samples, and a simulated feed-rate replay estimated a 1.7–1.9% throughput gain over a fixed design-feed policy. The historically evaluated microscope checkpoint reached IoU 0.870, 0.576 and 0.547 for pyrrhotite, chalcopyrite and pentlandite on 12 sections reused during development; the audited application build uses a different checkpoint, which showed large abundance errors on one section. We propose a six-month pilot at one host concentrator with both layers on site, the belt sensor leased and mounted only after a bench test on host ore, costed at R9.03–11.49 million and shared between the Technology Innovation Agency (TIA) and the host. Its targets, not yet results, are a phase-abundance error of $\leq 10\%$ on at least 90% of 30 locked host specimens and a hardness bound that holds on the host’s belt; if its gates pass, the microscope moves from technology readiness level (TRL) 4 to 5–6 and the belt system from TRL 3 to 6.

1 Introduction

As Team Sonar, working on Mintek’s Problem 3, our responsibility is to give a concentrator’s shift team a reliable reading of its ore before that ore reaches the mill, and to turn the reading into a plant setting that a person can approve. This report sets out what we built, how we tested it, what it would cost and earn, and, centrally, the six-month pilot that would test it on South African ore (Section 7). It describes a research prototype: the results are on public analogue data and a simulated plant. It is written for metallurgists, mineralogists, process engineers and the people who fund their technology. Technical terms are defined in the glossary (Table 15).

1.1 Problem statement

The brief states the problem plainly: “without real-time feedback, processing plants cannot adjust to changes in ore quality, leading to inefficient chemical usage and lower mineral yields” [1]. Ore reaches the mill within minutes of leaving the belt, whereas assays and automated mineralogy are batch laboratory work. Mintek’s own Minerals Processing and Characterisation Cluster runs QEMSCAN automated mineralogy [2], and a commercial laboratory describes mineralogy as “not an overnight assay service” [3]. Consequently, each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa, which mined an estimated 120 of the world’s 170 tonnes of platinum in 2025 (about 71%) and 70 of 190 tonnes of palladium. It also holds 63,000 of the world’s more than

76,000 tonnes of PGM reserves, mostly in the Bushveld Complex [4]. The same source attributes the 9% fall in South African PGM output in 2025 to lower palladium prices, higher deep-level mining costs and disruptions to the electricity supply.

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3 , and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [5]. Producers report recovery moves of this size to investors: Valterra raised concentrator recoveries by 1.2–1.7 percentage points at Mototolo and Amandelbult in 2025 [6]. Comminution, which sets both liberation and mill throughput, used 52% of site electricity in a benchmark of mines [7]. Plants do not decide entirely blind: operators, advanced process control (APC) and the mine plan already carry information about the ore. Thus, the decision we aim to improve is narrower: setting mill feed for each parcel of ore before it arrives, with a stated uncertainty, at a cost a producer can recover.

1.2 Proposed solution: three speeds, one decision

REEFPRINT/KHANYA reads the ore at three speeds (Section 3):

1. **the belt** (REEFPRINT), every 30 seconds: a short-wave infrared (SWIR) camera above the mill-feed conveyor feeds a model that predicts each parcel’s grinding hardness with an upper bound, and Bond’s law turns the bound into a mill

Table 1: The brief and the judging criteria: evidence and status. Shown = on public analogue data; Partial = part shown; Proposed = future work; Target = defined, not tested.

Requirement	Evidence (section)	Status
≥ 3 mineral phases	Pyrrhotite, chalcopyrite, pentlandite on analogue micrographs (4.4)	Shown
Accuracy report	Belt R^2 , bound coverage, baselines; microscope results per checkpoint (4.1–4.5)	Partial
$\leq 10\%$ error	Phase-abundance target defined; not yet tested on independent specimens (2.6)	Target
Adjust a plant parameter	Feed-rate proposal (designed); regrind advice to a simulator over OPC UA (2.3)	Partial
Real time	Hardness: 92 ms per parcel. Mineralogy: laboratory cadence, not on the belt	Partial
Local validation	Six-month pilot on host specimens and ore (7)	Proposed
Innovation	Bounded hardness input that abstains, with operator review (2)	Shown
Feasibility	Pilot plan, team, costs, TIA route, risks, TRL (7–8)	Proposed
Impact	Scenario value per 1% more metal (6.4)	Scenario
Technical execution	Out-of-fold tests, baselines, failed ideas reported; software tests by repository (App. A)	Shown

feed-rate proposal. The belt layer estimates hardness, not mineral identity;

2. **the laboratory** (KHANYA), once a shift: a microscope model classifies sulphide minerals in polished sections, which addresses the brief’s three-phase requirement on slower, prepared samples;
3. **the decision**: an operator reviews each proposed change before the ore reaches the mill; the site’s existing control system keeps authority, and every step is signed and recorded.

It is important to note what is not claimed: real-time mineralogy on the belt, performance on South African ore, a measured plant benefit, or identification of chromite and the silicates. Table 1 maps the brief and the judging criteria published in the hackathon FAQ (2025 edition; we have not seen a 2026 rubric) [8] to the evidence, with the status of each.

2 Methods

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data; none was supplied by Mintek or a producer. HIDSAG (CC0) supplies visible (39-band) and SWIR (75-band) scans of 146 Chilean porphyry copper–molybdenum drill-core composites with laboratory Bond work-index values [9]. A set of 1,112 Bushveld chromitite assay intervals (CC BY 4.0) was used to check PGE routing [10]. LumenStone S2 (research use only; 49 micrographs, 37 for training and 12 publisher test sections reused in development), a Norilsk nickel–copper sulphide ore [11], shares the mineral assemblage of Bushveld ores but not their abundance: sulphides cover 62.8% of its sections by area, whereas Merensky ore contains up to about 3% base-metal sulphides and UG2 less [5]. It supplies labels for the relevant phases, because the base-metal sulphides in UG2 are predominantly pentlandite, chalcopyrite, pyrrhotite and pyrite [5]; its much richer sulphide content and different preparation can still cause transfer failure, which only host specimens can test.

2.2 The belt layer: hardness from the infrared

Sensor (design). The design uses a SWIR line scanner such as the Specim SX25 (960–2500 nm, 640 pixels across; 162 lines a second assumed, to be confirmed for the configured unit) with a visible–near-infrared (VNIR) companion camera [12]. On a 1.2 m belt at 2 m/s it records a line every 12.3 mm with 1.88 mm pixels, so it measures parcels of ore, not particles. In the pilot design a *parcel* is 30 s of belt (about 60 m, or 4,860 scan lines averaged into one spectrum); in the replay reported here, a parcel is one HIDSAG composite scan of 80×117 pixels.

Model. Each parcel’s spectrum is reduced to features from the infrared absorption regions of clay, mica, chlorite and carbonate minerals (Fig. 2), and a model predicts the Bond work index (W_i , kWh/t). Results are reported only for samples the model never saw (out of fold), and a result counted only if it beat simple baselines under pass marks committed to the repository before each run. Five targets and several model variants were examined and only the work index survived, so the result is exploratory selection, not a confirmatory test; HIDSAG has no drill-hole identifiers, so grouped independence of the folds cannot be verified.

Bound and gate. The plant setting uses an upper bound that the true hardness should stay below 9 times out of 10 (a one-sided 90% split-conformal bound [13]), not the best guess. A more data-efficient bound [14] was tried first and failed its pass mark, so the exact bound that the plan named as its fallback is used, and the result is provisional. The 90% is a marginal coverage target that assumes new parcels resemble the calibration data; it is not a per-parcel guarantee and does not certify protection against overload. An out-of-distribution (OOD) check flags parcels unlike anything in training, and their advice is withdrawn.

2.3 The decision layer and its cadence

The decision layer turns the hardness bound into a mill setting that a person can approve. Ball mills are usually limited by motor power, so the tonnes a mill can grind depend on how hard the ore is. Bond’s law gives the specific grinding energy [15]

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \quad (1)$$

where F_{80} and P_{80} are the feed and product sizes (μm) passed by 80% of the material. At a fixed grind target the feed rate the mill can carry is proportional to $1/W_i$, so the proposal is

$$F_{\text{proposed}} = F_{\text{design}} \times \frac{W_{i,\text{design}}}{W_{i,\text{bound}}}. \quad (2)$$

The ratio removes explicit power and size terms only when power, feed size, product target and circuit efficiency are held constant; the pilot must test that approximation for the selected circuit. For example, a design hardness of 16.6 kWh/t and a bound of 16.9 kWh/t give 98.2% of design feed, inside an envelope set by the site’s metallurgist (85–110% in the demonstration). The demonstration falls back to 100% of design feed; at a site, the fallback is the host’s own operating procedure.

In the proposed cadence a person approves changes, not every computation: a parcel is scored as soon as it is scanned, and the operator has the rest of an assumed 90 s belt-to-mill transit to act. A proposal is computed every 30 s (at most 120 an hour), but the operator sees it only when it differs from the running setpoint by more than a deadband set by the metallurgist (for example 2% of design feed), and may approve,

modify within the envelope, reject or escalate. With no approval in time, a flagged parcel or stale data, the advice is withdrawn and the site’s own control policy continues; the system never raises feed on its own. Feed may be cut by up to 10% but raised by only 3% per parcel, and the shadow phase measures the real operator-facing rate before anyone acts. Every event is written to a hash-chained record with conventional (Ed25519) and post-quantum (ML-DSA-65 [16]) signatures; this supports traceability, not the correctness of the measurement. The microscope’s advice reaches a simulated regrind setting over OPC UA [17] (simulated); the feed-rate path is designed, not built.

2.4 The laboratory layer: minerals under the microscope

The microscope model is a standard image-segmentation network (DeepLabV3 with a ResNet-50 backbone [18, 19]) that labels every pixel of a reflected-light image as resin, pyrrhotite, chalcopyrite, pentlandite or magnetite. In the historical experiments, only the evaluated checkpoint (SHA-256 de7135a9; full hash in Appendix A) could send advice to the simulator. A *Quick* scan classifies six 512-pixel fields of a section; a *full* scan covers the whole section.

Connected regions of the mask are split into grains. For this demonstration, chalcopyrite and pentlandite are treated as payload and pyrrhotite as reject; that assignment is a demonstration assumption for this ore, because pyrrhotite’s value depends on PGM deportment and the host’s treatment objectives. In the historical experiments, a grain at least half payload by area was *payload-dominant* (“free”), otherwise *reject-dominant* (“locked”), a 2D proxy, not liberation. A rule-based advisor turned these measures into *grind finer*, *continue*, *verify before acting* or *adjust reagent*, as hypotheses for a metallurgist, answering *verify* within ± 0.335 of the 50% line (calibrated on the 12 evaluation sections, Section 4.5) and abstaining below 9 payload-bearing grains or confidence 0.85. The current application (region-aware contract version 3, measurement observed-target-support-v3) reports target-area categories instead, records retained target support and each exclusion cause, does not apply the historical margin, and always abstains from operational advice until a method-specific calibration exists.

2.5 Evaluation method

Belt models were scored out of fold against the strongest simple baseline (the fold mean, the fold median and a colour-only model), with pass marks committed to the repository before each run. The microscope was scored on the publisher’s 12 S2 test sections. They were excluded from network training but reused during development, to compare pipelines, select the checkpoint and calibrate the advisor band, so we report them as *development-influenced evaluation sections*, not an independent test. Accuracy is the intersection over union (IoU: the overlap between predicted and expert-labelled areas, from 0 to 1), pooled over 103.8 million pixels and averaged per section, with bootstrap confidence intervals over sections. Decision-level agreement runs the same advisor on the model’s mask and on the expert’s mask; a disagreement is *unsafe* if the model gives a confident instruction that the expert mask does not support. This measures sensitivity to segmentation error, not whether the rules themselves are metallurgically right.

2.6 Validation targets and independent protocol (proposed)

The $\leq 10\%$ error target must be defined before samples are collected, because absolute and relative errors differ: a reference of 2 wt% predicted as 3 wt% is a 1-point absolute error but a 50% relative error. We propose that the mineralogist and a statistician sign an acceptance contract naming the decision-critical phases, an absolute tolerance below an agreed quantification limit and a $\leq 10\%$ relative target above it, applied per specimen to every decision-critical phase (our proposed definition, Table 6). Image-area abundance is judged separately from decision calibration and from physical liberation. Microscope image fractions are 2D areas, whereas QEMSCAN modal results are usually mass fractions, and a density conversion alone does not make a polished surface represent bulk mass. So the comparison unit, aggregation and density assumptions are fixed in advance, and a same-surface area comparison comes before any mass estimate.

A blind test needs more than hiding answers: a separate custodian holds specimen identities and laboratory results; the frozen model receives only acquisition inputs and permitted metadata; predictions are hashed before the reference is revealed; splits are grouped by specimen and collection event; and every refusal stays in the denominator. We propose 30 development, 10 calibration and at least 30 locked specimens, subject to a power calculation and quotes; if a model fails, it is fixed on development material and must face a new untouched cohort. As of 7 October 2026 this protocol is drafted, not signed; Section 7.2 gives the pilot’s proposed targets.

3 System architecture

Figure 1 illustrates the system, highlighting that both branches share provenance and operator controls but have their own quality checks. The belt and laboratory layers run on different time scales: laboratory results give slower context and audit evidence, not a check on each 30 s parcel. The belt monitor (REEFPRIINT Live) runs the hardness model, the decision view and the signed record. The microscope workbench (KHANYA) is a web application (React) on a Python service (FastAPI and PyTorch) that runs on an ordinary laptop processor. In local mode it accepts connections only from the same machine and makes no external requests, so a network outage at the plant does not stop the advice. No language model computes a measured value or a control value. Each result records the input and checkpoint hashes, a versioned measurement recipe, the sampled-field coordinates and the share of the image actually observed, and is refused if any of these goes stale, so every number can be traced to what produced it.

Speeds were measured on a four-core laptop processor: 92 ms per parcel (median); 16.7–21.2 s per six-field microscope scan on mains power (5 runs) and 30.8 s on battery for the benchmark checkpoint; a fresh six-field scan in the audited build (fb78727d) took about 81 s on the audit machine, so the release chosen for the pilot will be re-benchmarked. A static public demonstration runs the belt monitor on precomputed data; a secure server passed 17 of 17 security tests locally and is not deployed. Software verification is in Appendix A.

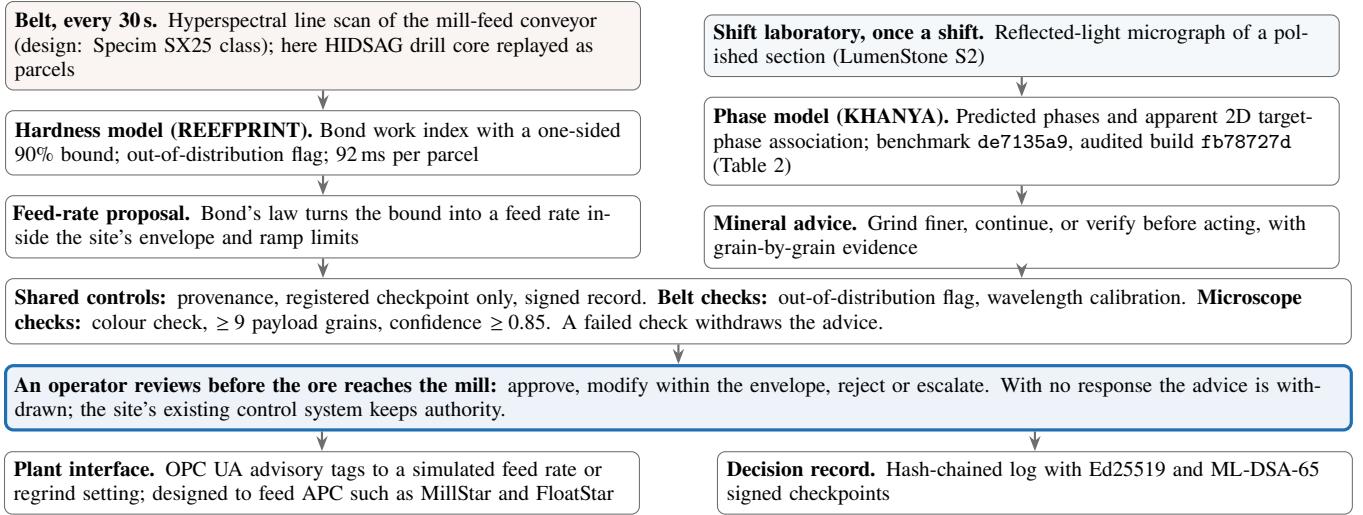


Figure 1: System architecture: three speeds, one decision. The belt and microscope branches produce model estimates at different cadences; the microscope gives predicted phase labels and apparent 2D target-phase association, and independent laboratory measurements provide the reference evidence. Shared provenance and operator controls are supplemented by branch-specific checks. Built: the models, applications and signed record (on replayed public data). Simulated: the plant and regrind interface. Designed: the belt sensor and feed-rate path.

4 Results

4.1 Belt: predicting grinding hardness

Of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beat its strongest baseline under every pass mark. Out of fold, the hyperspectral model explains 48% of the variance in work index (R^2 0.479, mean absolute error 1.05 kWh/t), a model given only colour explains 21% (R^2 0.210, 1.29 kWh/t), and the fold mean and median explain nothing (R^2 -0.027 and -0.041). Mean measured W_i is 14.62 kWh/t.

Figure 3 illustrates how the bound behaves, highlighting that it was exceeded on 10 of 146 samples (6.8%). The 95% interval for coverage (88.4–96.6%) spans the 90% target, so the result is compatible with it but does not establish future coverage, particularly on new ore. The price of the bound is width: the bound sits on average 1.64 kWh/t, about 11% of the mean hardness, above the best guess. The OOD check passed 137 parcels, marked 5 borderline and refused 4. For parcel GMET-0004 (Fig. 2) the bound of 16.9 kWh/t against a design hardness of 16.6 kWh/t (the training data’s 90th percentile) gives 98.2% of design feed; the laboratory value was 12.1 kWh/t, so the bound was conservative here.

4.2 Belt: from hardness to a plant decision

Figure 4 illustrates four feed-rate policies replayed over all 146 composites against a fixed design-feed policy (the 90th-percentile hardness), highlighting the trade-off between throughput and overloads. In the replay a parcel is *overloaded* when its true hardness exceeds the hardness the feed was set for; this is a model quantity, not a plant trip. Setting the feed from the bound, with OOD parcels withdrawn, gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%), and the overload share was 6.8% against 8.9%; the one-sided 95% upper bound on the difference (+1.4 points) is inside the +3-point margin in the evaluation plan. The original replay did not apply the 85–110% envelope; a later analysis with it gives +1.7% (+0.7 to +2.7%) and the same overload share. Feeding from the best guess instead would add 13.6% but overload 45.2% of parcels, which is why the policy uses a bound. Per-

fect knowledge would add 14.0%; the bound’s width is where the rest is lost, so a tighter bound from site data is the main route to a larger gain.

The result is provisional for three reasons: the exact bound was adopted after the more efficient one failed, HIDSAG has no time order (so transit time, stockpile mixing and the asymmetric ramp limit are not simulated), and the ore is not Bushveld ore. Over 200 random parcel orders, a $\pm 10\%$ ramp limit left overloads unchanged but $\pm 2\%$ added 4.7 points, so feed must be cut fast. A predefined robustness test found the error unchanged for a new capture, half the field of view, added noise and $\pm 15\%$ illumination, but 1.25 times larger, unflagged, under wavelength drift, so a wavelength-calibration check was added.

4.3 Physical plausibility checks

In the evaluated flotation scenarios, grinding faster cost up to 0.25 percentage points of recovery on the steep part of the recovery curve and none on the plateau; this is the largest modelled loss, not a physical bound, so the site’s curve must be measured. Re-ordering ore during load-shedding cannot create tonnes (it gave -3.8% in simulation), so that claim was rejected. The camera resolves parcels, not particles, and no test showed that it can identify minerals, so the microscope carries the phases.

4.4 Laboratory: phase identification

The microscope addresses the brief’s three-phase requirement on analogue ore. Figure 5 illustrates its accuracy by class, highlighting IoU values of 0.87 for pyrrhotite, 0.58 for chalcopyrite and 0.55 for pentlandite (precision and recall in Appendix A). The pooled mean IoU is 0.5725 (95% CI 0.494–0.624); averaged per section it is 0.4671 (0.412–0.534). The model clearly beats trivial baselines (majority class 0.117; colour-only 0.395).

Magnetite, a fourth class the brief does not require, is never predicted. It is only 0.8% of the test pixels: 78% of it is labelled as resin, 17% as pyrrhotite and 5% as pentlandite. The last share is about 0.3% of all pixels predicted as pentlandite, so magnetite barely moves the payload measures.

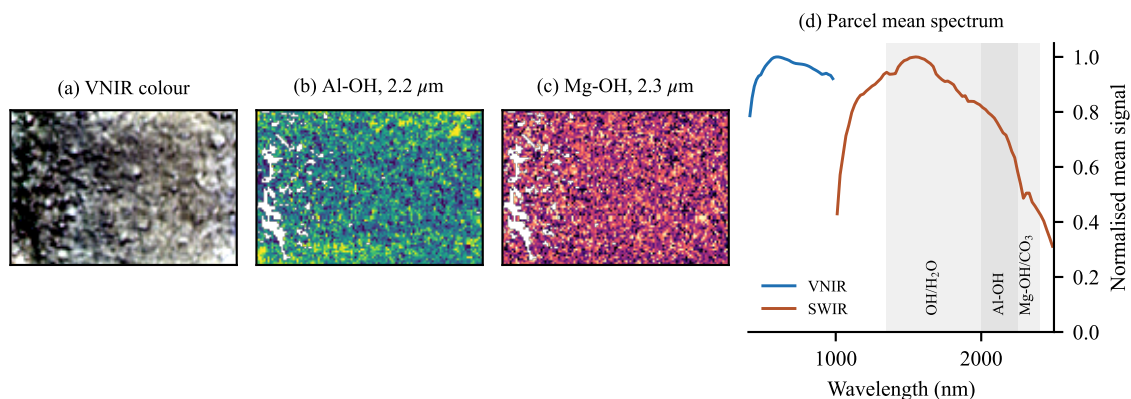


Figure 2: One belt parcel as the hardness model sees it: HIDSAG composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour image from the visible bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector stripes removed for display only. (d) Mean spectrum, with the absorption regions the model uses. The model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (laboratory value 12.1 kWh/t). The maps show absorption features, not a mineral identification.

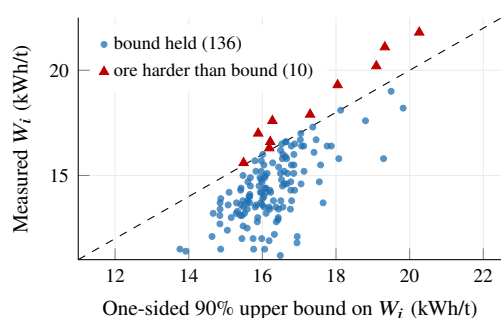


Figure 3: Calibration of the hardness bound, out of fold, 146 composites (no drill-hole grouping available). Points below the dashed line are samples the bound covered: 93.2% (95% CI 88.4–96.6%) against the 90% target. The bound sits on average 1.64 kWh/t (range 1.04–2.14) above the prediction.

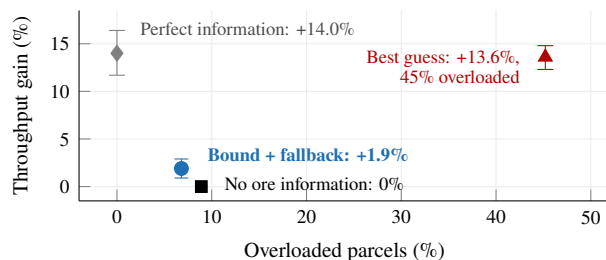


Figure 4: Feed-rate policies replayed over 146 composites against a fixed design-feed policy (simulated; not PGM ore, not a time series, original replay without the envelope), with 95% confidence intervals. Up is more throughput; left is fewer replay overloads.

Three checkpoints exist and must not share scores (Table 2). The figures above describe de7135a9. The application build audited on 6 October ran fb78727d, which the application itself marks as not approved. On one previously used section (test_11, six fields), the audit compared that model’s output with the expert masks: pixel accuracy was 64.1%, pyrrhotite was predicted at 61.1% of the area against 90.7% in the reference, and pentlandite at 34.8% against 2.0%. This is one development section, not QEMSCAN and not an independent cohort, but it is a concrete counterexample to any blanket $\leq 10\%$ claim for the active model. A retrained candidate, 42646cfa, scores 0.632 overall and detects magnetite, but loses pyrrhotite and pentlandite and has pentlandite precision of only 0.50 (Appendix A); it is quarantined, and the comparative deci-

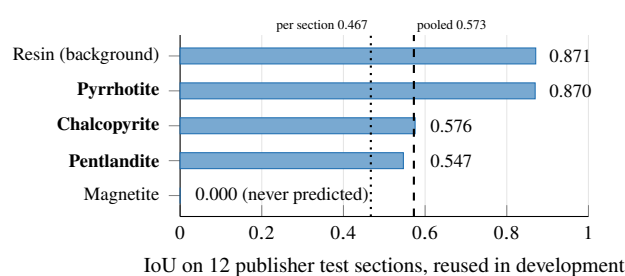


Figure 5: Microscope accuracy by class for the historically evaluated checkpoint de7135a9, full-section inference on development-influenced sections; the pooled mean is over all five classes. The three phases the brief requires are in bold. Per-section means range from 0.340 to 0.728.

Table 2: Model registry. Scores are bound to one checkpoint each. As of 7 October 2026, no checkpoint has been frozen for an independent test.

Checkpoint	Evidence	Status
de7135a9	Mean IoU 0.573 pooled, 12 development-influenced sections	Historical evaluated; research demo
fb78727d	Mean IoU 0.454 (re-run); audit on test_11: pixel accuracy 64.1%, pentlandite 34.8% vs 2.0%	Active in audited build; not approved
42646cfa	Mean IoU 0.632; pentlandite IoU 0.446, precision 0.50	Candidate; quarantined

sion safety of the checkpoints is untested. Two further continuations of it on six reused development sections (updating or freezing BatchNorm statistics) lowered pooled foreground IoU from 0.647 to 0.593 and 0.597 and raised pentlandite abundance error from 2.2 to 5.4 and 4.4 points, so both were rejected (continuation bundle of 7 October 2026, recomputed by an independent reviewer; not yet in the public repository).

4.5 Laboratory: decisions and grain evidence

We compared the historical advice from the model’s masks with that from the expert masks: of 8 sections where the expert mask said *grind finer*, the model agreed on 3, and of 3 *continue* sections it agreed on 1, abstaining on the rest. The model produced 4 definite recommendations, 6 *verify* outputs

and 2 too few grains outputs on the 12 sections, so its action coverage was 4 of 12 (33%). No definite recommendation disagreed in the unsafe direction with the same advisor applied to expert masks. This is a small, development-influenced rule-agreement result: it does not validate the rules themselves, establish an independent unsafe-action rate or demonstrate plant safety. The one-sided 95% upper bound of 22.1% on 0 of 12 refers to all sections, not to risk when the system acts. Both the rules and the abstention thresholds need independent evaluation on the host stream. Only 4 definite actions occurred, so even an ideal independent 0 of 4 would bound the failure rate only at about 52.7% (one-sided 95%). The counts are for de7135a9.

It is important to note that the ± 0.335 verify band was calibrated on these same sections, so the agreement result is partly in-sample. On the 9 sections the payload rule governs, the result stays at 0 wrong for bands of ± 0.25 or wider, but one section becomes unsafe below that and two with no band, so the band must be re-set on independent pilot sections. A realistic exposure shift (-35 levels on each colour channel) changed the advice on 8 of 12 sections, so illumination control is also a pilot requirement.

The current application implements the relabelling and a border policy: edge-touching grains are excluded and counted. A development audit of six reused parent images (24 Quick and full rows; supplied continuation evidence, re-checked by an independent reviewer, not yet public) found that even the expert reference masks keep only 0–5.8% of observed target pixels in Quick mode and 0–70.5% in full mode, so none meets the provisional 80% support policy and all 24 rows abstain. This points to the component-as-particle assumption and specimen preparation, not only to segmentation, and phase-area estimation may remain useful where particle association is not.

5 Comparison with literature and limitations

5.1 Comparison with literature

Published work supports the plausibility of the approach, not its validity here. Notole et al. [20] validated hyperspectral imaging and portable XRF against reference methods on Merensky Reef drill core scanned at the Council for Geoscience in Pretoria, with long-wave infrared as well as VNIR/SWIR sensing; that shows the sensing works on Bushveld material, not that our VNIR/SWIR hardness model does, and it is not a public paired conveyor dataset. The three sulphides the microscope classifies are the main base-metal sulphides of UG2 ore [5], and our 50% proxy line coincides with a boundary between middling classes in Mintek’s liberation scheme [2], although that boundary does not by itself define free material. Mintek’s AI optimisation raised a lead–zinc concentrator’s throughput by 0.5%, worth about R33 million a year [21], so moves of this order are valued.

Four ideas were tested and withdrawn, and they mark where the method stops. The belt spectra did not predict QEMSCAN mineralogy once size fraction and process line were accounted for; a grouped re-analysis (99 records, 36 composites) found only 23–91% of eligible records within 10% relative error, and a process-line baseline beat the spectral model for chalcopyrite and pyrite, so the belt claim is limited to hardness. Belt chemistry routed Bushveld PGE ore worse than the mine plan’s seam label (balanced accuracy 0.79 against 0.82); the belt did not predict recovery, lime or pH; and plant tags never

beat the last assay at forecasting concentrate silica.

5.2 Limitations

The following limitations apply, and (a), (b), (e) and (h) block any investment decision until tested: (a) no South African ore has been evaluated, and a model trained on Chilean porphyry core may not transfer, because hardness also depends on silicification, grain size and fracturing; (b) the belt result is a replay of 146 composites without time order or drill-hole grouping, against a fixed design-feed policy rather than the plant’s existing control, and a camera sees the belt surface, which may not represent the mass entering the mill; (c) the belt camera predicts hardness and does not identify minerals, and the microscope has not been trained on chromite or the silicates; (d) the 12 microscope sections guided development, and the historical checkpoint cannot be regenerated exactly; (e) the advisor’s rules (including the pyrrhotite-reject and 50% assumptions) are unvalidated for any host ore, its band was set on the evaluation sections, and payload-dominance is a 2D proxy, not liberation; (f) latency is measured on one laptop, the plant is a simulator, and economic benefit has not been attributed at any site; and (g) there is no fresh paired PGM validation, the active application checkpoint differs from the evaluated one, chromite and silicate classes are missing, and reference-laboratory uncertainty is not yet quantified; and (h) the latest local build records retained target support and exclusion causes, binds results to its startup build identity and withholds operational advice; the earlier excluded-support counterexample now abstains and the historical margin is not applied, but inadequate support even in reference masks leaves specimen preparation, the component-as-particle assumption and method-specific calibration unresolved. Since the accepted abstract, the five Bushveld phases became three sulphides on an analogue ore, and the fallback became the host’s own procedure (the demonstration’s 100% design feed can raise feed, so it is not universally conservative).

6 Business model and financial case

6.1 Value proposition and business model

Our value proposition is simple: the plant gets a reading of its ore before the mill, with a stated bound, and beyond a fixed support fee it pays a success fee only on value that an independent party has measured. No producer, Mintek or TIA has agreed to fund, host or license the work; the roles below are proposals. The buyer is the concentrator or metallurgy manager at a South African PGM producer, who owns recovery, concentrate quality and throughput; named producers are market segments, not customers. Figure 6 illustrates how money and value flow, highlighting that only the success fee depends on measured value. TIA, the national agency that funds technology towards commercialisation [22], is the proposed main funder (Section 7.5).

The proposed commercial structure has three parts: a capped validation contract; a service fee after acceptance; and an optional variable fee capped at 20% of independently verified net value, with the baseline, measurement period, adjustments, negative-value treatment and audit rights defined in advance. The success fee is zero when verified net value is zero or negative; validation charges and the support fee are separate, so no blanket zero-fee promise is made. Willingness to pay is untested, and laboratory savings are not claimed. The

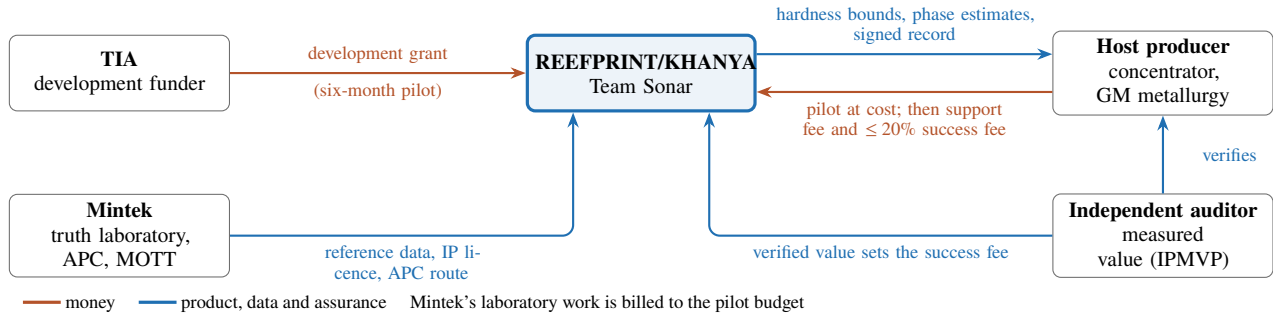


Figure 6: The proposed business model; partner roles are proposals, not agreements. The pilot would be paid at cost; after a successful trial the producer would pay a fixed support fee and a success fee capped at 20% of independently verified net value, which is zero when that value is zero or negative. Mintek would earn from the laboratory work the models need; licensing follows the hackathon IP agreement with its Office of Technology Transfer (MOTT).

model fits Mintek's published work [21]: the models need laboratory work, and Mintek's benefit-sharing model with Hindustan Zinc has generated over R35 million since 2021. Valterra's realised basket price rose from R32,611 per ounce in 2025 to R45,993 in the first half of 2026 [6, 23].

Plants already have cross-belt neutron analysers (bulk elements, with isotope sources that need a licence [24]), slurry analysers (grade after milling), automated mineralogy (phases, in days) and APC such as MillStar and FloatStar. The system replaces none of them; whether it adds ore-side information beyond APC, operator practice and mine-plan blending is what the pilot must show.

6.2 Pilot budget and cash release

Table 4 sets out one scoped planning allowance for the six-month pilot, in two tracks: R4.93 million for the laboratory and team track and R4.10–6.56 million for the belt track. It replaces the earlier first tranche and separate belt stage; the earlier R8.2–49.2 million installation grid survives only as a superseded sensitivity in Table 5. Staff rates are team planning rates; development specimens are referenced by expert annotation, so only 44 go to automated mineralogy; and every line is replaced by a local quote at G0, with the question narrowed or returned for a new decision if quotes exceed it. The sensor is leased or borrowed with a purchase option where possible, so that a failed bench gate (G1B) returns it before installation; the belt allowance is a purchase-equivalent ceiling. VAT, duties, major civil work, conveyors or sorters, shutdown costs and closed-loop control are excluded, so the totals are scoped budgets, not worst-case losses. After the pilot, annual support is assumed at US\$30,000–60,000 (R0.49–0.98 million).

The two tracks have independent tranches (Table 3), so a belt failure cannot stop the laboratory work. The 20 bench Bond tests (about R0.39 million) needed before G1B are in tranche L1, committed at G0. If G1B fails, B2 is never released: the belt track stops having spent B1 plus any lease cancellation fee, and the laboratory track continues. Committed laboratory orders, staff notice periods and lease cancellation are the costs a stop cannot recover; the payer of each line is agreed at G0.

6.3 From throughput to recovered metal

The belt result is a throughput gain, and revenue comes from recovered metal, so the two must be bridged. For a consistent stream, metal output is proportional to dry tonnes × head

Table 3: Cash release by track (planning allowances; payer agreed at G0).

Tranche	Released at	Pays for	Rand
L1	G0	Staff and site, months 1–2; imaging kit; inductions; first sections; 20 bench Bond tests	1.13 M
L2	Week-8 laboratory review (not G1B)	Remaining staff, site, sections, automated mineralogy, 24 belt Bond tests, review	3.15 M
B1	G0	Sensor lease deposit or bench hire; edge computer	0.57–0.90 M
B2	G1B, week 8	Installation, rest of sensor, belt sampling, vendor support	2.71–4.35 M
Reserves	Documented risk event	15% laboratory, 25% belt contingency	1.46–1.95 M
Total			9.03–11.49 M

grade × recovery, so

$$\frac{\Delta M}{M} = \left(1 + \frac{\Delta T}{T}\right) \left(1 + \frac{\Delta G}{G}\right) \left(1 + \frac{\Delta R}{R}\right) - 1, \quad (3)$$

where T is throughput, G head grade and R recovery. With unchanged grade, a 1.75% throughput gain and a recovery loss of 0.25 percentage points from 87% (a relative change of −0.29%) give 1.46% more metal. Where the plant is limited by ore supply, extra throughput is worth nothing. If all of Zondereinde's output passed the instrumented circuit (an upper bound), 1.46% more metal would be worth R64–123 million a year in margin, and a quarter of that gain R16–31 million; these are simulated gains, not measurements.

6.4 Financial methodology

The following assumptions apply: (a) annual output is that of Northam's Zondereinde mine, 333,050 equivalent refined 4E ounces in F2026 [28]; (b) prices are Valterra's realised basket prices of R32,611 per ounce (2025) and R45,993 (first half of 2026), with cash operating costs of R19,488 and R20,677 per ounce [6, 23], used as proxies for another company's mine, with different baskets, payability and cost structure; (c) for a commercial-life case, installed cost is the belt allowance without the six-month support line (US\$175,000–280,000, R2.87–4.59 million) and running cost is the post-trial support (R0.49–0.98 million a year); (d) the equipment has a 5-year life at a 10% discount rate; and (e) an affected-output fraction f of the mine's output passes the instrumented circuit, taken as 1 unless stated. The annual value of an uplift u in recovered metal

Table 4: One scoped budget for the six-month pilot: team planning allowances, not quotations. US\$ list prices are public anchors from one university facility and one industry guide; rand at R16.4 per US\$ (January–September 2026 average [25]). Every line is replaced by a quote at G0.

Laboratory and team track			Belt track: one belt on site	
Item	Rand	Basis	Item	US\$ (rand)
Two pilot engineers (the authors)	480,000	2 × 6 months × R40,000	Sensor, optics, references (lease or purchase-equivalent)	100–160k (R1.64–2.62 M)
Software and OT-security engineer	360,000	To recruit; 6 months × R60,000	Lighting, enclosure, mounting	25–40k (R0.41–0.66 M)
Independent mineralogist	280,000	35 days × R8,000; annotates development specimens	Edge computer, UPS, network	10–15k (R0.16–0.25 M)
Statistician and data custodian	144,000	18 days × R8,000	Installation, integration, commissioning	40–65k (R0.66–1.07 M)
Site living, travel, medicals, inductions	318,000	3 people × 6 months; PPE	Belt sampling crew, references, vendor support	25–40k (R0.41–0.66 M)
Polished-section preparation	154,000	77 sections × R2,000	25% contingency	50–80k (R0.82–1.31 M)
Automated mineralogy	1,443,000	44 specimens × US\$2,000 list [26]	Belt track total	250–400k (R4.10–6.56 M)
Bond ball-mill work index	866,000	44 composites × up to US\$1,200 [27]		
Microscope imaging kit	120,000	Allowance		
Independent review, IP and data terms, close-out	120,000	Allowance		
Contingency (15%)	643,000	Released only on a documented risk event		
Laboratory and team total	4,928,000	No overrun without a new decision	Pilot total, both tracks	R9.03–11.49 M

is

$$V = u \times f \times Q \times m, \quad (4)$$

where Q is annual output in ounces and m the margin per ounce. For $u = 1\%$ and $m = \text{price less cash cost}$,

$$V = 0.01 \times 333,050 \times (\text{R}13,123 \text{ to } \text{R}25,316), \quad (5)$$

which is equivalent to R43.7–84.3 million a year (R108.6–153.2 million in revenue before costs). Deducting the full average cash cost may be conservative where the mill has spare capacity, but extra mining, handling or bottleneck costs could change that, so a host-approved incremental cash-flow model is needed before investment. The installed cost is spread over the equipment’s life with the capital recovery factor

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = 0.2638 \quad (r = 10\%, n = 5), \quad (6)$$

so the annualised cost is

$$C = \text{CRF} \times C_{\text{installed}} + C_{\text{running}}, \quad (7)$$

which is equivalent to R1.25–2.20 million a year, or R2.50 million with the 25% contingency on the high case. The break-even uplift is then $u_{\text{BE}} = C/(f \times Q \times m)$. Table 5 shows how it moves with cost, margin, affected fraction and the fee basis. If the 20% fee is charged on the margin before system costs, the producer breaks even only when $0.8V = C$. The break-even figures are far smaller than the precision with which a short plant trial can measure uplift, which is why measured value, not break-even, decides G5.

Annualising over five years describes continued successful use, not whether a six-month pilot pays for itself. The pilot cash view is simpler: R9.03–11.49 million is a scoped planning allowance, not a maximum exposure, because VAT, duties, shutdown and civil costs are excluded and must be assigned to a payer. It is released in tranches (Table 3), so a stop at any gate leaves later tranches unspent; if the pilot runs to the end and the host declines to continue, the full allowance is spent, with equipment resale assumed to be zero in the pessimistic case. The fee basis must be agreed so that no support

Table 5: Break-even sensitivities in relative recovered metal (recalculated under the stated assumptions; illustrative, not a site funding threshold). One percentage point of recovery at 80% recovery is 1.25% relative.

Scenario	More metal needed
Low cost, 2026 margin, all output affected	0.015%
High cost, 2025 margin, all output	0.050%
High cost + 25% contingency, 2025 margin	0.057%
High cost, 2025 margin, 20% fee before system costs	0.063%
High cost, 2025 margin, half of output affected	0.10%
High cost, 2025 margin, a quarter of output affected	0.20%
High cost, 30% lower basket price (2025 costs)	0.20%
Superseded R49.2 M installation grid, 2025 margin	0.52%

cost is charged twice, and measured value needs an agreed metallurgical accounting boundary and attribution method under IPMVP [29].

7 The six-month pilot

As Team Sonar, our pilot problem is this: in six months at one host concentrator, find out whether the microscope meets a $\leq 10\%$ phase-abundance target on independent local specimens, and whether the hardness bound holds on the host’s own belt, in shadow and then in a short supervised advisory window. Both layers run on site. A clear negative answer at any gate is a useful result, because it stops spending early.

7.1 Pilot duration and feasibility

The following assumptions make six months realistic: (a) contracting is finished before month 0 at gate G0, including a scoped, quoted measurement-feasibility review of specimen preparation and image–reference correspondence, a leased or demonstration sensor with a confirmed delivery date and a planned plant stop in weeks 8–10, so the clock starts at a readiness event, not a funding announcement; (b) the sensor

Table 6: Proposed pilot targets. “Agreed” values are signed with the host and statistician before evaluation data are collected.

Target	Pass condition (proposed)	Gate
Phase abundance	Pentlandite, chalcopyrite, pyrrhotite: $\leq 10\%$ relative error above an agreed reporting limit, absolute tolerance below it; ≥ 27 of 30 locked specimens pass, lower bound reported	G1A
Coverage	Refusals and failures stay in the denominator; answered share $\geq$ agreed minimum	G1A
Bench transfer	20 host composites (development and calibration of the bench method): error below a metadata baseline; agreed bound width	G1B
Belt shadow	24 paired belt composites as locked evaluation of a frozen version, with no online recalibration: error below baseline; predefined count inside the bound, interval method, maximum width and refusal rate; scoring inside the transit time	G3
Advisory window	Proposals withdrawn when they should be; no setpoint outside the envelope; operator response logged	G4

is non-contact and mounts above an existing conveyor without modifying it, with read-only data, so installation is a job of days within that stop, not a construction project; (c) specimens come from the host’s existing shift sampling and belt cuts, so 70 microscope specimens and 44 Bond composites over about 16 weeks need only about 5 and 3 a week; (d) reference results return in fortnightly batches within an agreed three weeks; and (e) the software, model registry and abundance-error checks already exist, and the latest build already abstains where support is inadequate, so month 1 goes on measurement feasibility, local data and development-only method selection; a calibration is fitted later, on its own cohort, after the pipeline is frozen. As of 7 October 2026 no host, laboratory or sensor access is arranged, so these agreements, the reference method and the error definitions are G0 prerequisites, and $\leq 10\%$ remains a target. Two limits apply: the sensor is mounted only if it passes a bench test on host samples (G1B, week 8), and the four-week advisory window can show the workflow, timing and operator response, but cannot measure a gain of about 2% with useful precision, so independently measured value (G5) needs a longer trial after the pilot. If supply, delivery or the plant stop slips, the pilot is extended; the cohort is never cut to keep the date.

7.2 Pilot objectives and acceptance targets

Table 6 sets out the proposed targets, signed with the host before any evaluation specimen is collected. We use “90% accuracy” in one strict sense for the microscope: at least 90% of locked specimens must meet the error contract for every decision-critical phase. With 30 locked specimens, an observed 27 of 30 shows only that the true pass rate exceeds 76% (one-sided 95%); demonstrating 90% itself needs 29 of 29, or 45 of 46 if one failure is allowed, and these bounds assume independent specimens. So 27 of 30 is a proposed observed acceptance rule, not proof of a 90% pass rate; 46 specimens would add about R0.56 million before contingency. On the belt, 24 composites give exploratory evidence of limited precision, and because a bound can cover 90% simply by being too wide, its width and refusal rate are judged with its coverage.

Table 7: The pilot team (functions; one person may cover several, except the independent ones).

Role	Time	During the pilot
Pilot lead, field and systems (S. Khumalo)	Full time, on site	Sampling, custody, imaging, sensor checks, reporting
Model and evaluation lead (L. Hoaeane)	Full time, on site	Model freeze, abundance contract, belt-transfer and shadow analysis
Software and OT-security engineer	To recruit; full time	Sensor integration, offline operation, read-only network, signed record
Independent mineralogist	About 35 days	Annotates development specimens, reviews references
Statistician and data custodian	About 18 days	Blind split, sealed predictions, power check, evaluation
Vendor commissioning engineer	In the belt allowance	Sensor installation, calibration, site acceptance
Host metallurgist, instrument technician, operators, safety and OT officers	Host, in kind	Use case, sampling, envelope, lockout, inductions, advisory reviews, sign-off
Reference laboratory	Quoted	Sections, automated mineralogy, Bond tests

7.3 Work plan and daily site routine

Figure 7 illustrates the plan, highlighting that the sensor is tested on host ore before it is mounted, and that the microscope model is frozen before any locked reference is revealed. A typical day on site would run as follows (proposed; the host sets the hours and safety rules):

- (i) shift handover and safety talk; check the sensor’s health log, white and dark references, window and purge air; review the overnight parcel log and refusals;
- (ii) collect sampler cuts and belt cuts, each time-stamped to its scanned parcel, and log them in the custody register; split for sections, Bond composites and retention; the custodian’s script assigns each specimen to development, calibration or locked;
- (iii) image returned sections under controlled light with a colour reference; locked outputs are hashed and sealed before any reference is seen;
- (iv) development work on development data only (calibration specimens are used once, after the freeze); compare shadow predictions with arriving Bond results; in the advisory window, sit with the control-room operator at each proposal review;
- (v) a short review with the host metallurgist; back up and sign the day’s records.

Weekly, a reference batch is dispatched; monthly, a steering meeting checks spend, risks and schedule.

7.4 Pilot team

Table 7 lists the people the pilot needs. The two authors are in place and both work on site for the full six months; the software and operational-technology (OT) security engineer must be recruited before G0, and the independent roles are contracted, so that assessment never collapses into self-certification. The host owns plant operation and stop authority; in shadow nothing is written to the plant, and in the advisory window only an operator can change a setpoint.

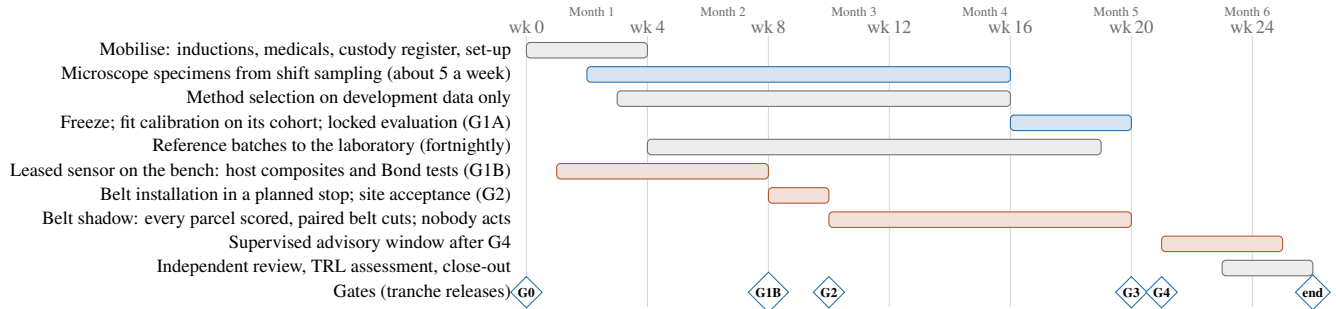


Figure 7: The six-month pilot, in weeks from the readiness event (planning estimate). Blue bars: microscope track; copper bars: belt track. G0 releases L1 and B1; the week-8 laboratory review releases L2; G1B releases B2 (Table 3). G1A falls in week 20 with G3. If G1B fails, the leased sensor is returned before installation and the microscope track continues.

Table 8: Proposed funding structure around TIA (proposals; nothing is agreed).

Stage	TRL	TIA instrument	Cash and other parties
Pilot: laboratory and team	4 to 5–6	Technology Development Fund	TIA R3.93 M; host pays Bond tests (R1.00 M with contingency) and gives site access, staff time and laboratory space; Mintek reference work at cost
Pilot: belt	3 to 6	Development Fund; Matching Fund only if eligible	R4.10–6.56 M; team proposes 50:50 with the host (R2.05–3.28 M each), not an agency term
Microscope product	5–6	Seed Fund, terms of the current call	To be confirmed; technology-transfer office
Value trial and roll-out	7–9	Matching or Commercialisation Support	Host-led; investors

7.5 Funding structure and the role of TIA

TIA funds technology from proof of concept to pre-commercialisation, with instruments matched to readiness [30]. Its Technology Development Fund lists TRL 4–7 and funds demonstration and pilot plants and techno-economic studies, so it is the route we propose to investigate for this pilot. Its Seed Fund is listed for TRL 3–8 [30]; one published Seed Fund call, run by the SAMRC for its own staff, sets TRL 5–6 and up to R1.5 million, and asks for a functional prototype, a roadmap, an industry partner and an IP strategy [31]. That shows what an application needs, not our eligibility. The Industry Matching Fund is described for late-stage technologies [30], so it is not assumed for the TRL 3 belt model. Beyond money, TIA links innovators with technology-transfer offices, incubators and investors. The applicable instrument, eligibility, cap and co-funding terms will be confirmed with TIA and a university technology-transfer office for the current call; Table 8 is a team proposal, with no application made and no amount, split or equipment ownership agreed.

7.6 Pilot risks and mitigations

Table 9 sets out the risks of the pilot period, as the team assesses them, each with a mitigation before it happens and a fallback if it does. Contingency pays only for uncertain costs inside the approved scope. Three fallbacks run through all of them: the host’s controls keep authority, each stage ends at a gate where spending can stop, and no success fee is due without measured value.

7.7 Governance, compliance and training

These are planned controls, not achieved compliance (Table 14). The Mine Health and Safety Act requires a risk assessment with the health and safety committee, and training before significant changes to plant [33], which applies to the belt installation; POPIA governs approvers’ names in the record [34]; Mining Charter clause 2.3.1 counts skills and R&D spend [35]; IP from public funding falls under the IPR Act [36], although the contract settles ownership; and Mintek and TIA spend under the PFMA [37]. Independent checks are outsourced to a SANAS-accredited laboratory [38], Mintek’s QEMSCAN, a value auditor using a plan adapted from IP-MVP [29], an IEC 62443 assessor, an IP attorney through MOTT and a university model reviewer. Each group is trained before its phase, aligned with the site’s skills plan [39]: laboratory staff and metallurgists before G1A work starts, instrument technicians at installation, and control-room operators before G4; competence is shown by observed tasks and sign-off, not attendance.

8 TRL milestones and gates

Table 10 places each component on the standard readiness scale [40] as a provisional self-assessment. A level is assessed by the independent reviewer from demonstrated performance against agreed criteria, not awarded automatically at a gate or from elapsed time. The belt system climbs furthest in the pilot, because it moves from public core to the host’s own belt in two steps: host samples on the bench (TRL 4, G1B) and shadow operation on the belt (TRL 6, G3). The advisory window starts operational evidence towards TRL 7; independently measured financial value (G5) is a separate commercial gate, not a TRL criterion.

Table 11 lists what each gate must show, who decides, and what happens if it fails. After the pilot, an extended advisory trial with on/off operating blocks gives the independently measured value for G5, so a commercial decision is about 12 months from the readiness event.

9 Conclusions and recommendation

The present contribution is a functioning, inspectable evidence workflow and a falsifiable route to local validation. On public analogue data the hardness model improves on simple baselines and the microscope classifies three relevant sulphide phases, but the feed-rate benefit is a simulation, the microscope results are development-influenced, the active applica-

Table 9: Pilot-period risk register: mitigation before the event, and the fallback if it happens.

#	Risk during the pilot	Mitigation (before)	Fallback (if it happens)
P1	Fewer than 5 specimens a week (stoppages, sampler faults)	Sampling plan agreed at G0; collect across shifts; weekly count	Extend collection; never shrink the locked cohort
P2	Laboratory turnaround over 3 weeks, or quotes above budget	Turnaround written into the quote; fortnightly batches; second laboratory pre-qualified	Narrow the question or seek a new decision; no silent cuts to repeats
P3	Sensor delivered late, or no lease or demonstration unit	Delivery date and lease terms at G0; two vendors approached	Shift the belt track; the microscope track continues
P4	Planned plant stop missed for installation	Installation booked with the host at G0; mount pre-assembled and tested on the bench	Install at the next stop; shorten the advisory window, not shadow
P5	The sensor fails the bench test (G1B)	Bench test on host composites before mounting	Return the sensor; stop the belt track; spend only tranche A
P6	Dust, moisture, vibration or wavelength drift on the belt	Enclosure and purge air; references every cycle; wavelength check	Refuse affected parcels; clean or re-calibrate
P7	Scan and belt sample do not match in time	Time-stamped parcels; belt-cut timing checked against transit time	Exclude unmatched pairs and report them
P8	Operator over-trust or alarm fatigue in the advisory window	Deadband; training before G4; every proposal reviewed with an engineer	Advice expires; host control continues; end the window early
P9	Unsafe write to the plant or cyber incident	Read-only shadow; advisory tags only; IEC 62443 zones [32]; IT/OT sign-off	Disconnect; the plant runs on its own controls
P10	The microscope misses the $\leq 10\%$ target, or data leak between sets	One frozen model; independent custodian; sealed predictions	Report the negative result; collect a fresh cohort before any new claim
P11	Engineer not recruited, host withdraws, or data and IP terms unsigned	Recruit and sign before G0; named deputies; LumenStone rights clarified	Hold the next tranche; keep lawful evidence; qualify another host
P12	Rand 10% weaker on US\$-priced lines; safety incident at the belt	Rand quotes where possible; lockout, escort and permits near moving plant	About R0.64–0.89 M, met from contingency; stop work and follow host procedure

Table 10: Provisional self-assessed TRL now, and the evidence that could support each next level.

Component	Now	End of pilot, if criteria are met	After
Microscope workflow	4: validated in the laboratory on public analogue data	5: independent host specimens (G1A); 6: shift routine	7–8
Belt hardness model	3: proof of concept on public core; simulated plant	4: host samples on the bench (G1B); 6: shadow on the host belt (G3)	7
Decision layer and record	4: built; replayed data; simulated plant	6: shadow; 7 begun in the advisory window (G4)	7–8

Table 11: Gates, decision owners and stop outcomes. Numerical thresholds are signed before evaluation data are collected.

Gate	Release condition	Decides	If it fails
G0	Host letter, quotes, sensor lease and delivery, plant stop booked, data and IP terms, signed contract	Funder, host	Preparation only
G1B	Hardness transfers to host composites on the bench (week 8)	Reviewer, host	Return sensor
G2	Risk assessment, read-only design, site acceptance (week 10)	Host OT, safety	Repair or remove
G1A	Microscope targets met on locked specimens (week 20)	Reviewer, host	Image review only
G3	Belt shadow targets: coverage, timing, drift, refusals (week 20)	Metallurgist	Extend or stop
G4	Envelope, change management, training, rollback (week 21)	Change board	No advisory window
G5	Independently measured net value and total cost (after the pilot)	Auditor, host	No success fee

tion checkpoint has shown large abundance errors on one section, and the $\leq 10\%$ error target is untested. We therefore recommend a six-month pilot at one host concentrator that runs the microscope in the shift laboratory and a leased sensor on the host's belt, costed at R9.03–11.49 million and shared be-

tween TIA and the host in gated tranches, with the $\leq 10\%$ error target tested on at least 30 locked host specimens and the sensor mounted only after it passes a bench test on host ore. As of this submission, the independent local phase-abundance target has not been demonstrated; a commercial claim will depend on independent reference agreement, useful acceptance coverage and measured net value on the affected stream.

CRedit authorship contribution statement. **Lethabo Hoaeane:** conceptualisation, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. **Sibusiso Khumalo:** microscope model and evaluation, advisor and safety gates, OPC UA simulation, verification, technical report.

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Data and code availability. HIDSAG is available under CC0 [9]; LumenStone under research-use terms [11]; the Bushveld assays under CC BY 4.0 [10]. Code, checkpoint hashes and results files for the public revision: <https://github.com/Sibusiso-K/KHANYA> (Appendix A); the latest continuation evidence is a local, manifest-bound bundle, not yet public. A static demonstration of the belt monitor runs at <https://lethabomh14-reefprint.static.hf.space>.

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Table 12: Microscope per-class results, historical checkpoint against the re-trained candidate, 12 S2 publisher test sections reused in development (full-section inference).

Class	Historical de7135a9			Candidate IoU (precision)
	IoU	Recall	Precision	
Resin	0.871	0.920	0.942	0.848 (0.960)
Chalcopryrite	0.576	0.716	0.746	0.800 (0.937)
Magnetite	0.000	0.000	—	0.248 (0.255)
Pyrrhotite	0.870	0.934	0.926	0.818 (0.950)
Pentlandite	0.547	0.742	0.675	0.446 (0.503)
Mean	0.573			0.632

Table 14: Laws, policies and standards, and the planned controls (not achieved compliance).

Law, policy or standard	What it asks, and the planned control
Mine Health and Safety Act 29 of 1996 [33]	Risk assessment with the health and safety committee and training before plant changes (ss. 10, 11, 34): site risk assessment, lockout, trained operators
Protection of Personal Information Act 4 of 2013 [34]	Limited processing of approvers' names: purpose, access, retention and non-disciplinary use agreed with the information officer
Mining Charter 2018, clause 2.3.1 [35]	5% of the leviable amount on skills, including R&D: pilot training and R&D may count
IPR from Publicly Financed R&D Act 51 of 2008 [36]	IP identified and protected for South Africa: managed by MOTT under the hackathon agreement [8], subject to the contract
Public Finance Management Act 1 of 1999 [37]	Mintek (Schedule 3B) and TIA (3A) spend under PFMA controls: tranches tied to gates, auditable deliverables

Table 13: Pixel confusion for the historical checkpoint (% of each true class), 103.8 million pixels.

True \ predicted	Resin	Ccp	Mag	Po	Pn
Resin	92.0	0.8	0.0	5.7	1.5
Chalcopryrite	1.6	71.6	0.0	3.1	23.7
Magnetite	78.3	0.1	0.0	16.8	4.8
Pyrrhotite	1.1	1.6	0.0	93.4	3.9
Pentlandite	0.7	0.8	0.0	24.3	74.2

Table 15: Glossary.

APC	Advanced process control, e.g. Mintek's MillStar and FloatStar
Bond work index (W_i)	Energy needed to grind ore, in kWh/t; harder ore has a higher W_i
Bound	A hardness the true value should stay below 9 times in 10
Fallback	Demonstration: 100% of design feed; at a site, the host's own procedure
Payload-dominant	A grain at least half payload by 2D area ("free" in the app); a proxy, not liberation
TRL	Technology readiness level, 1 (basic principles) to 9 (proven in operation) [40]
PGM, 4E	Platinum-group metals; 4E = Pt, Pd, Rh and Au

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A Supporting results

Microscope: Tables 12–13. **Tests** (counts, not a performance metric). Public revision 85061d4: workbench 181 unit-test functions and 7 browser scenarios; belt software (separate repository) 328. Local continuation package (7 October, manifest-bound): 321 passed, 4 skipped (supplied); 59 overlapping tests re-run by a reviewer passed. **Checkpoints** (*SHA-256*). Historical

de7135a96541a46dc0981a991cb954186c7cd669ea1cb33914d1929ae9b1357; audited fb78727d4859947d3605ccf9374f8defbc40897e9cf1b1922a5c21832d387067; candidate 42646cfafbeac5398368dba17f7ad3531ffcea572e6342cc4fcb42ed0b44b3b. **Reproducibility.** Inference reproduces from archived weights; retraining does not (unseeded rerun 0.454; seeded since adeb1ee). **Smoke checks:** 52.9 s, then 72.9 s on one real image; zero retained support; control held.